Write Advantages and Disadvantages of Database management System.

• Advantages of Database management System:

i) Data Redundancy Reduction:
Avoids duplication of data by Storing it centrally.

ii) Data Consistancy:
Ensures the Same data appears Correctly in all places.

iii) Data Security
provides access control, authentication, and authorization.

iv) Data integrity:
maintains accuracy and Control of data using Constraints.

v) Data Sharing
multiple user Can access data Simultaneously.

vi) Backup and Recovery.
protects data from System failure or Crashes.

• Disadvantages of Database management System:

i) High cost
Software, hardware, and maintenence and expensive

ii) Complexity.
Requires Skilled professional to design and manage

iii) Large Size
DBMS Software needs more storage space.

iv) Security Risk:
centralized data can be targated through cyber attacks.

v) maintenance Cost
Regular updates, backups and tuning required
vi) Single point failure.
If DBMS fails, entire System will get affected

Q2 Explain difference between file processing and DBMS.

file processing System	DBMS.
i) Data is Stored in Seperate files.	i) Data is Stored in centralized database.
ii) High data Redundancy	ii) minimal data Redundancy.
iii) Data inconsistancy possible	iii) Data consistancy maintained.
iv) poor data Security.	iv) Storage data Security mechanism.
v) difficult data Sharing	v) easy data Sharing among users.
vi) No data integrity Constraints	Integrity Constraints enforced.
vii) No backup & Recovery Support.	vii) Automatic Backup & Recovery.

viii) program data dependence.	Data Independence.
ix) difficult to modify structure.	ix) easy to modify database structure.
x) Suitable for small applications	Suitable for large and complex applications.
xi) No query language for processing.	xi) uses language like SQL for processing.

Q3 Explain Database language in detail with suitable example.

Database language are used to define, store, Retrive, Control and manage data in a database. In DBMS languages are mainly classified into 4 types.

i) DDL - Data defination language.
ii) DML - Data mainpulation language.
iii) TCL - Transiction Control language.
iv) DCL - Data Control language.

i) DDL : used to define and modify the Structure of data base objects like table, schema, View, indexes.

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Common DDL Commands :
• Create • ALTER, DROP, TRUNCATE, RENAME.
• example :

```
CREATE TABLE Student (
        Rollno INT PRIMARY KEY,
        Name VARCHAR (50);
        Age INT,
        Course VARCHAR(30) );
  ALTER TABLE Student ADD Email VARCHAR
        (50);
```

ii) DML :
Used to insert, update, delete, and Retrive data from database tables.
Common DML Commands :
• INSERT, • UPDATE, • DELETE, • SELECT.
example :

```
• INSERT INTO Student VALUES (1, 'Radhe', '19',
       'Computer');
• UPDATE Student SET Age = 20 WHERE Rollno:
       1;
• DELETE FROM Student WHERE Rollno=1;

• SELECT * From Student;
```

iii) DCL :
used to control access and permissions.
Common DCL commands :
• GRANT, • REVOKE

• example :

GRANT SELECT, INSERT ON Student To user1;
REVOKE INSERT ON Student From user1;

iv) TCL :
used to manage transactions and ensure data consistency.
Common TCL Commands :
• COMMIT , • ROLLBACK, • SAVEPOINT.

example :
SAVEPOINT Sp1;
INSERT INTO Students VALUES (2, 'Madhav', 19, 'IT');
ROLLBACK TO Sp1;
COMMIT;

Q4 Construct an ER diagram for travel agency. Consider various entities such as travel agency, passanger, Branch, Seat, bus, employee, tours, etc. Specialization and Genralization EER feature.

• Specialization and Genralization EER feature :
- Specialization of employee:
employee is speciaized into :
a) manager
- manager-ID
- experience
- Bonus

b) Driver
- licence-NO
- Driving-Experience.

c) clerk
- Shift - Timing.

• Type :
- Disjoint Specialization
- partial Specialization.

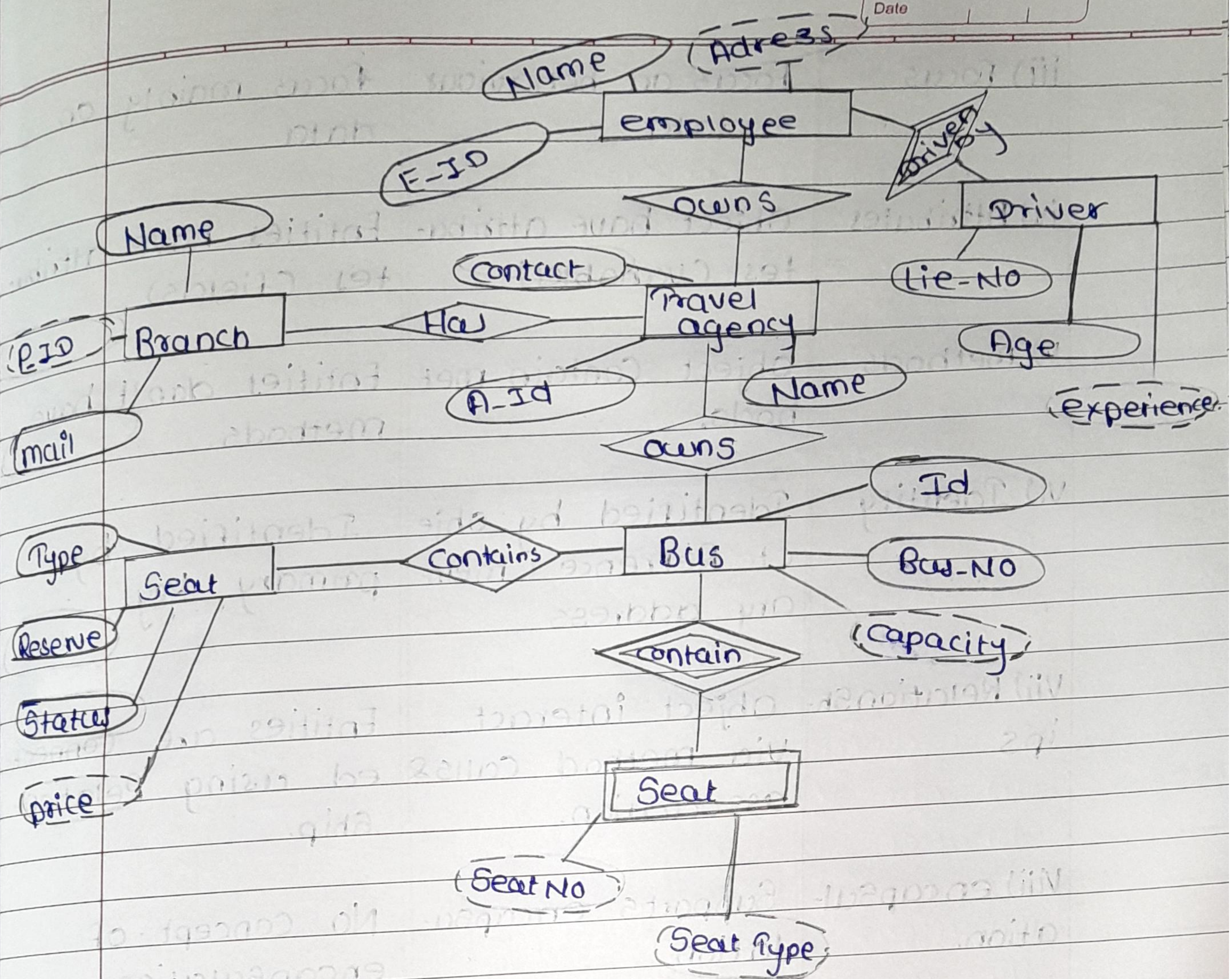
• Generalization Of Tour
Tour
Specialize into :
a) Domestic - Tour
- State
- city
b) International - Tour
- Country
- Visa-Required.

• Type
- Overlapping allowed
- Total Specialization.

• Enhanced ER Constraints Summary :

Specialization	Employee -> manager, Driver
generalization	Tour -> Domestic, International
Disjoint	employee Specialization
Overlapping	Tour Specialization
Total participation	Tour
partial participation	Employee.

• ER Diagram :



Q5 Comparision of Object (oop) and 'Entity' in ER model.

Aspect	Object (OOP)	Entity (ER model)
i) defination	An object is a instance of a class in OOP.	An entity is a real-world object Represented in a database.
ii) purpose	used for Software design & Implementation.	used for database designing & data modeling.

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iii) Focus	focus on behaviour + data	focus mainly on data.
iv) Attributes	Object have attributes (variables)	Entities have attributes (fields)
v) methods	Object Contain methods.	Entities don't have methods.
vi) Identity	Identified by object Refrence / memory address	Identified by primary key.
vii) Relationships	Object interact via method calls & association.	Entities are Connected using Relationship.
viii) encapsulation.	Supports encapsulation	No concept of encapsulation.
ix) Inheritance	Supported through class inheritance	Supported through genralization / Specialization (EER).

Q6 who is a DBA ? what are the Responsibilities of DBA ? also explain what are the components of Storage manager ?

A DBA database administrator is a person Responsible for managing, Controlling, and maintaining the database System.

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The DBA ensure that the database is Secure, avaliable, efficient, and reliable.

- Responsibilities of a DBA :
  - i) Database design
  - ii) user Authorization & Security.
  - iii) Backup and Recovery.
  - iv) Performance monitoring & Tuning.
  - v) Data integrity
  - vi) Concurrency Control.
  - vii) Storage management
  - Viii) Database maintenance

- Components of Storage manager :

The Storage manager is a DBMS Components that manages the Storage of data on disk and provides an interface between low-level data Storage and high-level queries.

i) Authorization and integrity manager.
ii) Transaction manager
iii) File manager
iv) Buffer manager
v) Recovery manager.

Component		function.
Authorization manager	-	Access Control
Transaction manager	-	manages transactio
file manager	-	disk file mng.
Buffer manager	-	memory-disk transfe
Recovery manager	-	failure Recovery.

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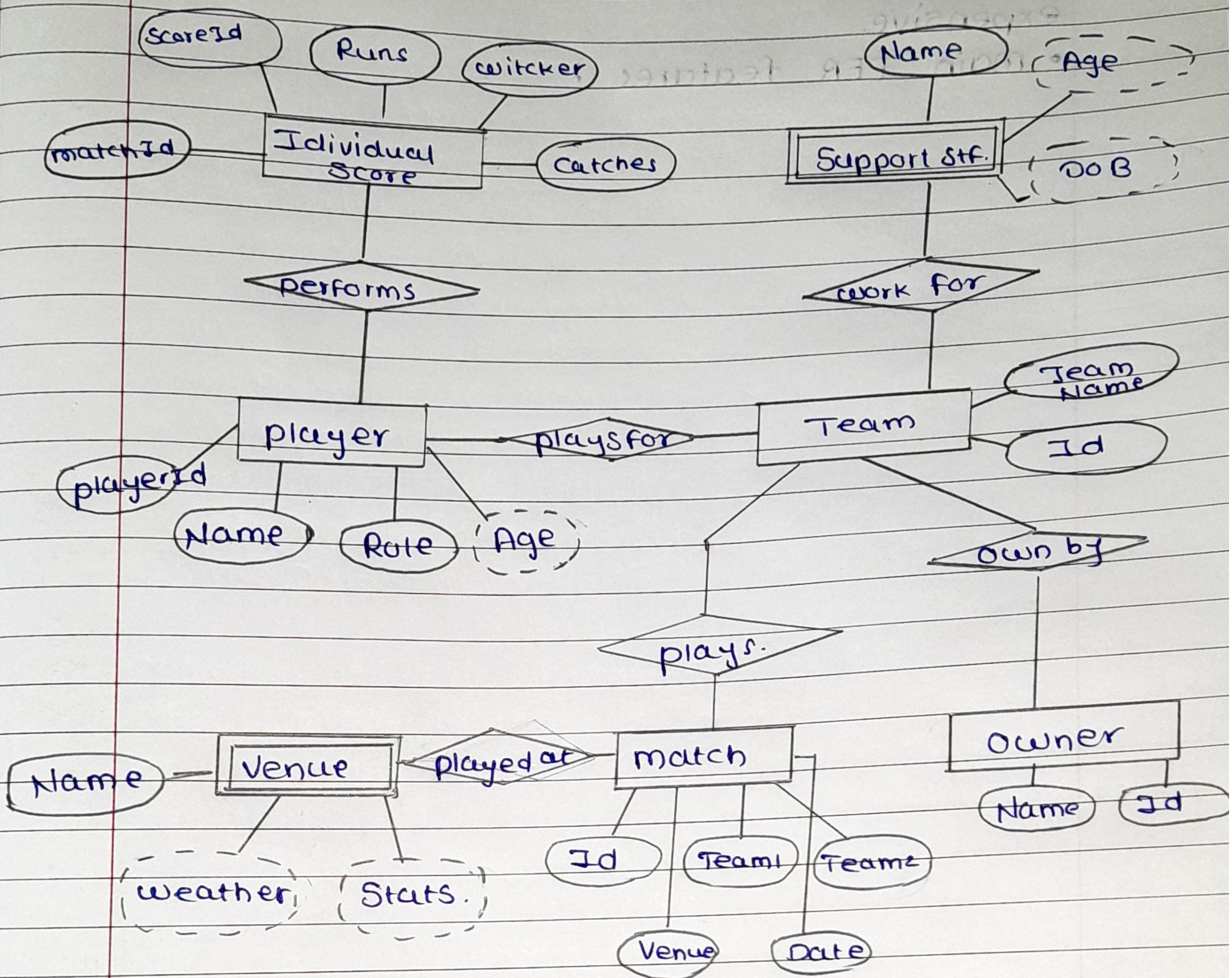
Q 7 what are the terminologies which is helpful to draw a complete ER diagram? Draw a ER diagram on indian premier league, as per your Requirements.

- Terminologies Helpful to draw an ER diagram :

Terminology	Representation
i) Entity	Rectangle
ii) Relationship	Diamond
iii) Attribute	Oval
iv) Composite attribute	Oval
v) derived attribute	dotted oval
vi) Cardinality	
a) one to one	1 | 1
b) one to many	1 | N
c) many to many	N | N
vii) weak Relation	
viii) weak entity	

Thus, The proper utilization of these terminologies will Result in having a meaningful ER diagram.

- ER Diagram for IPL :



Q8 What is E-R extended features in DBMS ? Explain In Brief ?

• Extended ER model is an enhancement of th basic ER model.
It is used to Represent more Complex real-world Situations in database design.
ER extended features add advanced concepts like inheritance, hierchy, and constraints to th ER model making it more powerful and